Nathan Zhao

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Education

Georgia Institute of Technology Atlanta, Georgia

B.S. in Electrical Engineering (Circuit Technology, Robotics) *GPA: 4.0 · Expected December 2028*

Relevant Coursework: Digital Design Fundamentals, Circuit Analysis, Intro Signal Processing, Linear Algebra, Multivariable Calculus, Differential Equations, Physics (Mechanics & Electromagnetism)

Activities: Nanotechnology & Photonics Society (Equipment Engineering) · Propulsive Landers (Avionics)

Relevant Skills

Design & Simulation: KiCad, Altium Designer, LTSpice, PTC Onshape, SolidWorks, AutoCAD (P&ID)

Lab and Fabrication: Digital Multimeter, Oscilloscope, Logic Analyzer, Soldering (Through-hole & SMT), Probe Station, Chemical Vapor Deposition, Physical Vapor Deposition, Metrology/Characterization, FDM/SLA 3D Printing

Embedded & Firmware: C/C++, PlatformIO, PicoSDK, I2C, SPI, UART, CAN, EtherCAT, BLE, ROS/ROS2

Software and Machine Learning: Python, C#/.NET, Java, MATLAB, PyTorch, OpenCV, Linux, Docker, Git

Experience

Founding Engineer · Effluent 2026

- Designed and fabricated custom sensor-array carrier PCBs in KiCad, developed C++ microcontroller firmware interfacing with digital sensors over I2C and bridging telemetry to analytics systems with MQTT/open62541.
- Built simulation tools, Python data pipelines, and a cross-platform C#/.NET GUI/HMI to visualize ML scenario modeling insights and real-time sensor analytics for enhanced WWTP process control.
- Engineered time-series machine learning forecasting models for continuous sensor data streams, focusing on robust edge feature extraction and transfer learning across facilities.

Research & Development Intern · MRIGlobal 2025 – 2026

- Led hardware development of portable, low-cost functionalized gas sensor arrays with STM32 for real-time hazardous chemical classification, executing schematic capture and PCB layout in Altium Designer.
- Built sensor functionalization tools, a vacuum-oven vapor-deposition system, and sensor-characterization test rigs with closed-loop analyte mass-flow-control software via RS-232 communication and a custom Python HMI.
- Deployed optimized CNN-BiLSTM embedded models (TensorFlow Lite) on-device, achieving >96% preliminary classification accuracy on real-time edge inference for gas signatures.

Research Assistant · UMKC Multiplatform Interactive Robotics Lab (mirolab) 2025 – 2026

- Developed wearable hand-joint goniometry system using Kalman fusion of multi-view hand pose estimation with a custom FDM-printed ArUco glove, achieving sub-degree error in flexion-angle measurement validation.
- Designed the wearable's TPU kirigami/living-hinge flexures and a stepper-driven characterization rig for repeatable sensor validation; co-authoring manuscript intended for submission to *IEEE RA-L* (PI: Hee Sup Shin).

Research Intern · University of Pittsburgh, Accessible SmartTech Lab 2024

- Developed autonomous object-handoff behaviors on a Hello Robot Stretch RE1 (ROS Noetic) using ArUco markers, YOLOv5 object detection, and 2D LiDAR SLAM (PI: Dan Ding).

Selected Projects

Bimanual Mobile Manipulator for Assisted Living ↗ 2024 – 2026

Regeneron ISEF 2025 4th Grand Award, GKCSEF 2025 Grand Winner

- Designed and built a dual-telescopic-arm mobile robot with custom 3D-printed wave-drive actuators, electromechanical architecture, power distribution, and embedded firmware.
- Implemented PIDF actuator control with quadrature encoder feedback for millimeter-scale arm positioning.

mmHome: Multimodal Contactless IoT Vital-Sign Monitoring & Fall Detection ↗ 2025 – 2026

MIT THINK 2026 Semifinalist, GKCSEF 2026 Pioneer in Science

- Designed and built a low-cost ambient sensing platform integrating 24GHz mmWave radar, piezoelectric vibration sensors, ADCs, and MEMS IMUs with an ESP32; developed microcontroller firmware and analog acquisition circuitry, applying Savitzky-Golay filtering and signal feature extraction for ML-based detection.